

Design and Implementation of a Database Management System for Event Registration and Venue Scheduling System in University

Problem statement

Manual event management breaks down:

- Venues get double-booked
- Participants register twice for the same event
- Attendance isn't tracked reliably
- Certificates get issued without verifying eligibility

Our solution

A relational database that automates event setup, session scheduling, venue allocation, participant registration, attendance, payments, and certificate & report generation — end to end.

Objectives

1

No venue overlap

No two sessions may share a venue at overlapping times

2

Capacity limits

Registrations can't exceed a venue's seating capacity

3

No duplicate registration

One participant, one registration per event

4

Attendance-based certificates

Eligibility computed from attendance threshold

5

Non-negative payments

Every recorded payment amount is validated ≥ 0

6

Operational reporting

Seats, registrations, attendance, revenue, feedback

Users & scope

Who interacts with the system, and what it covers

Admin / Coordinator

Creates events, schedules sessions,
allocates venues, manages the system

Speaker

Assigned to sessions; schedule
generated as a report

Participant

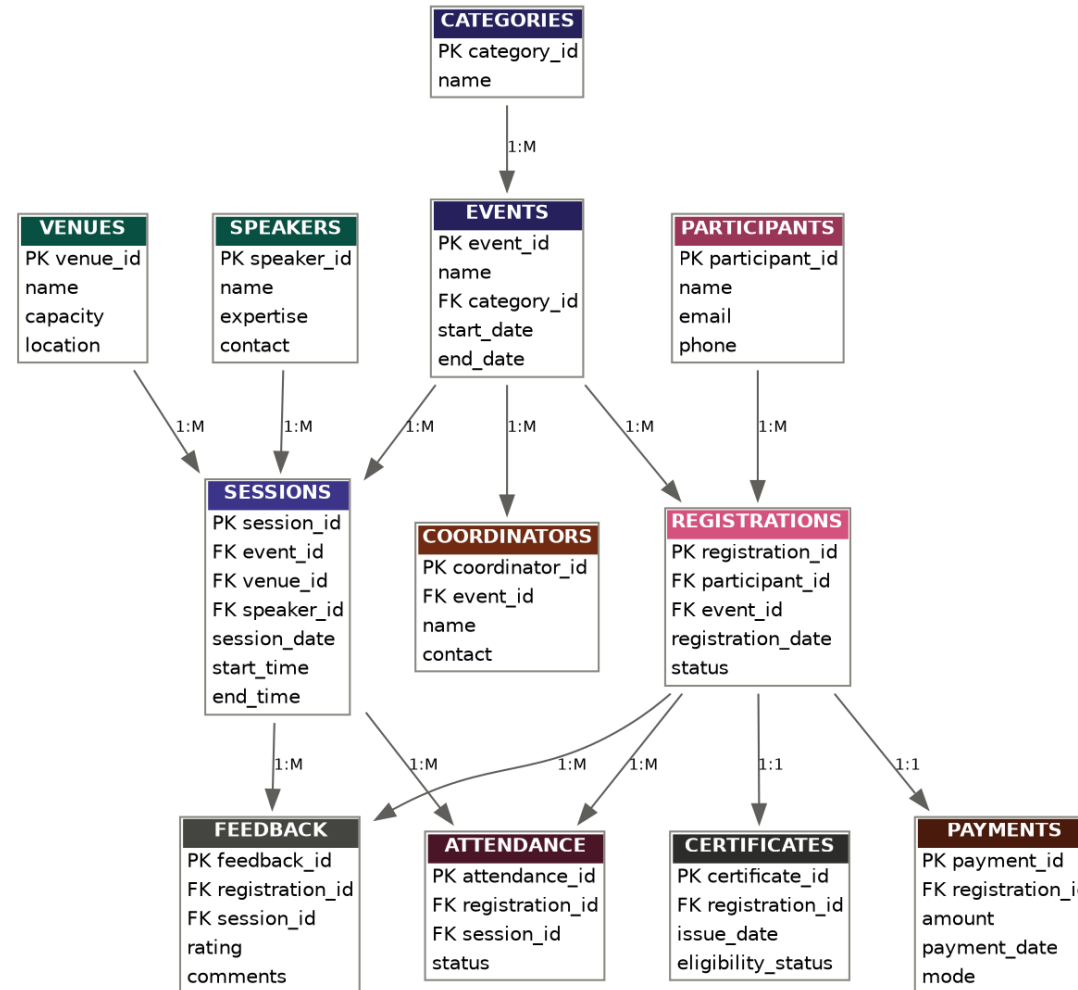
Registers for events, tracked for
attendance, receives certificates

Scope

Event creation & categorisation · session scheduling · venue allocation · registration · payments · attendance · certificate eligibility · feedback · reporting

Entity-relationship diagram

12 entities modelling the full event lifecycle



Key relationships

- 1 Category (1) → Events (M)
- 2 Event (1) → Sessions (M) | Venue (1) → Sessions (M) | Speaker (1) → Sessions (M)
- 3 Event (1) → Coordinators (M)
- 4 Participant (1) → Registrations (M) | Event (1) → Registrations (M)
- 5 Registration (1) → Payment (0 or 1)
- 6 Registration (1) → Attendance (M) | Session (1) → Attendance (M)
- 7 Registration (1) → Certificate (0 or 1)
- 8 Registration (1) → Feedback (M) | Session (1) → Feedback (M)

Initial relational schema

registrations	payments	attendance	certificates
registration_id PK	payment_id PK	attendance_id PK	certificate_id PK
participant_id, event_id FK	registration_id FK, amount	registration_id, session_id FK	registration_id FK, UNIQUE
Rule: UNIQUE(participant_id, event_id)	Rule: CHECK amount >= 0	Rule: status: present / absent	Rule: eligibility_status

Business rules → database constraints

No duplicate registration

```
UNIQUE (participant_id, event_id)
```

Non-negative payments

```
CHECK (amount >= 0)
```

Venue overlap prevention

```
Time-range validation on venue_id  
+ session times
```

Attendance-based eligibility

```
eligibility_status derived from % present
```

Thank You !!